

Quantum state spaces of nonrigid molecules and their complexes

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Abstract

Cooling, trapping and coherent control make molecular configuration states useful objects in their own right, not only tools for calculating spectra. We separate the molecular state-space structure from its coordinate realizations and effective models, making their symmetry actions, choices and approximation assumptions explicit. Starting from the full centered nuclear configuration space, we extend the quantum theory of molecular orientation [1] beyond a rigid rotational orbit to internal rearrangements and molecular complexes. We give a spectral criterion for candidate regimes and bounds on omitted couplings, explicit symmetry actions on compatible local coordinate descriptions, exact level counts and spin weights within the version model, and chart-free Gaussian packets with calculable species content.

Part I states the model, builds the regime from the bottom (the point group from a reference geometry, then the group closed under the resolved motions, inside the interval of candidates) and computes its exact outputs. For any candidate subgroup in the interval, a compatible reference family gives an explicit action on local coordinates (Part II), inherited from the linear action of S on configuration space. Figure 1 shows the whole chain, and one molecule, methanol, runs through both parts.

The rigid setting of molecular orientation [1], large-amplitude reference paths [2, 3], internal-axis frames [4, 5] and mechanical connections [6, 7] each treat part of this structure in a setting of their own. We organize these coordinate descriptions around a common ambient configuration space and carry its permutation–inversion action explicitly through the local charts and their transitions, so that it is visible how the configuration action appears in local coordinates, how patches and frame transitions fit together, and where effective-model assumptions enter.

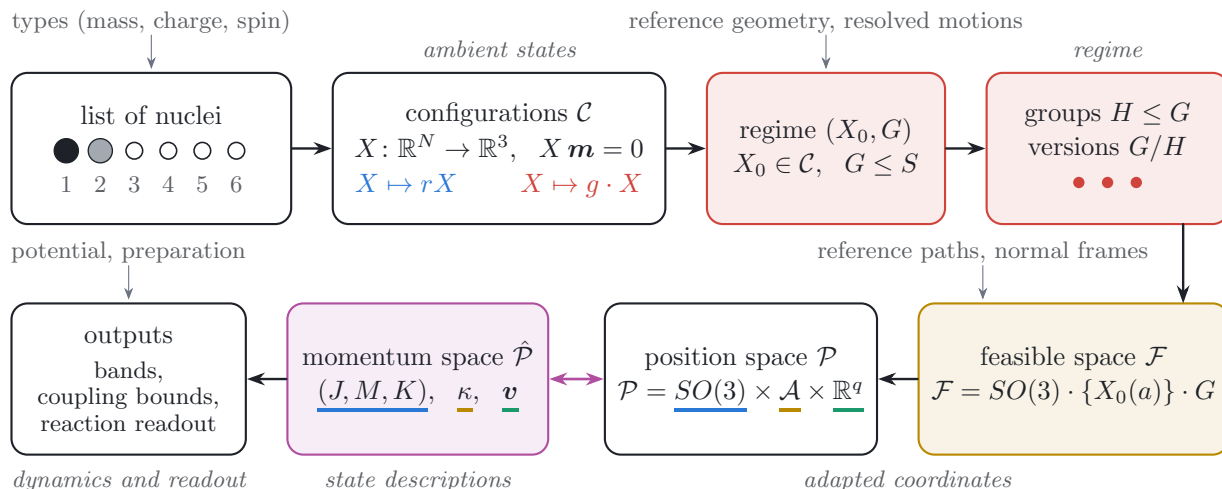


Figure 1: The construction in five layers (*italic*): ambient states, the regime, adapted coordinates, state descriptions, and dynamics with readout. Grey arrows are the inputs each layer needs; the pink double arrow is the Fourier transform between position and momentum labels, whose coloured bars pair each factor of $\mathcal{P}$ with its label in $\hat{\mathcal{P}}$. The dots are methanol’s six nuclei and three versions.

AI disclosure. The author worked with Claude (Anthropic), ChatGPT (OpenAI), and Gemini (Google) in developing the theory, writing the supporting scripts, and drafting this paper.

Part I. The model and its exact outputs

1 The space

Definition 1 (Labels and configurations). Number the nuclei $1, \dots, N$. Let $\mathbf{e}_1, \dots, \mathbf{e}_N$ be the standard basis of $\mathbb{R}^N$. Nucleus i has a type: mass m_i , charge and spin I_i ; nuclei of one type are *like*. Write $\mathbf{m} = \sum_i m_i \mathbf{e}_i$. A *configuration* is a linear map $X : \mathbb{R}^N \rightarrow \mathbb{R}^3$ with

$$X \mathbf{m} = 0.$$

The column $X \mathbf{e}_i$ is the position of nucleus i (Figure 2), and the constraint puts the centre of mass at the origin. The rows are the components along the lab axes. Axis labels are upright: x, y, z for the lab axes, x', y', z' for the body axes. The configurations form the vector space $\mathcal{C}$ of dimension $3N - 3$.

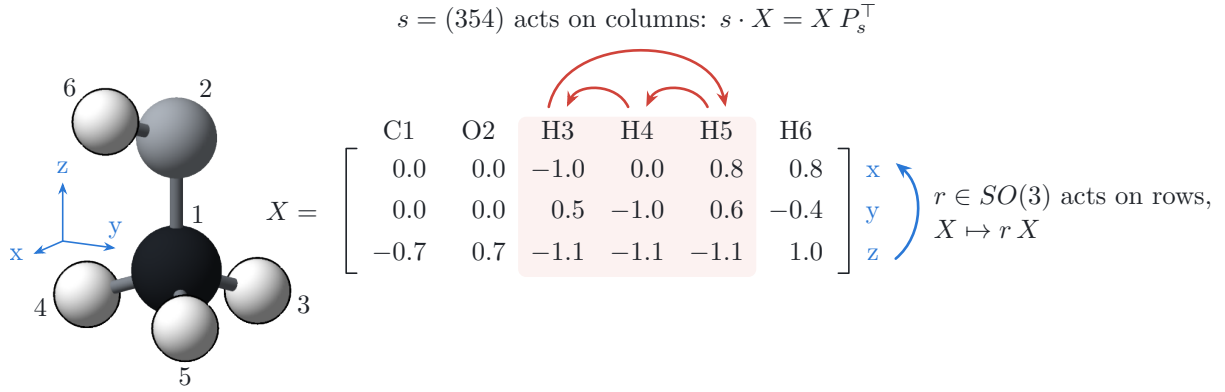


Figure 2: Methanol (left) and its configuration X (right), a 3×6 matrix whose column i is the position of nucleus i in Å, rounded for display. Rotations act on the rows, the lab axes x, y, z , and relabelings on the columns: $s = (354)$ cycles the methyl columns H3, H5, H4.

Definition 2 (The two actions). A rotation $r \in SO(3)$ acts on the left, $X \mapsto rX$. A *relabeling* g permutes like labels, with or without the inversion of space. Its inversion sign $\det g$ is -1 if it inverts and $+1$ otherwise, and it acts by multiplying the matrix from the right,

$$g \cdot X = (\det g) X P_g^\top,$$

with P_g its permutation matrix. The relabelings form the finite group

$$S = \left(\prod_{\text{types}} S_{n_{\text{type}}} \right) \times \{E, E^*\},$$

with E the identity and E^* the inversion alone; inverting elements are marked with a star.¹

The two actions are on opposite sides of a matrix, so they commute:

$$g \cdot (rX) = r(g \cdot X).$$

The inversion $X \mapsto -X$ could sit on either side. It sits in S , so rotations are proper and $SO(3) \cap S$ is trivial, and it must: a feasible group can contain $(23)^*$ without (23) or E^* , as ammonia's does when its umbrella motion is not resolved.

¹ S is the complete nuclear permutation inversion group of Longuet-Higgins [8] and of Bunker and Jensen [9].

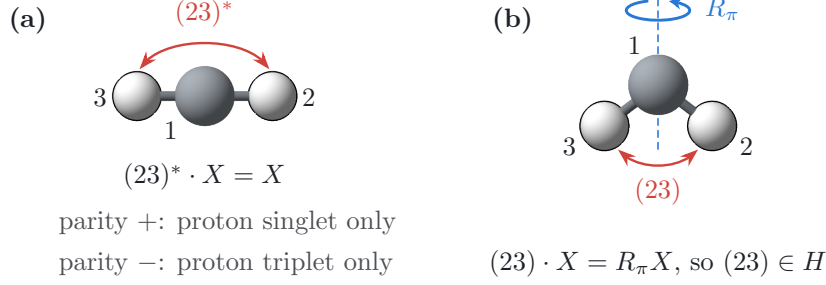


Figure 3: (a) Relabeling and inverting straight water returns the same configuration, $(23)^* \cdot X = X$, so the node rule (2) leaves only the proton singlet there at parity + and only the triplet at parity -. (b) In bent water (23) gives the copy turned by the half turn R_π about the dashed axis, so (23) lies in the point group H . A straight arrangement is a regular point of $\mathcal{C}$; its axial stabilizer is handled by the chart of Section 5, not by singular coordinates.

Free space and traps. Without the constraint $X \mathbf{m} = 0$ configurations fill $\mathbb{R}^{3N}$, which the mass metric splits orthogonally into the centre of mass $\mathbf{R}_{\text{cm}} \in \mathbb{R}^3$ and $\mathcal{C}$. Rotations turn $\mathbf{R}_{\text{cm}}$, and a relabeling sends it to $(\det g) \mathbf{R}_{\text{cm}}$. In free space the kinetic energy separates, states lie in $L^2(\mathbb{R}^3) \otimes L^2(\mathcal{C})$, and everything below concerns the second factor. Translations commute with unstarred relabelings, a starred one reverses them, and a rotation turns the translation vector; parity labels refer to the rest frame. In a trap, centre-of-mass and internal motion need not separate: the potential may depend on $\mathbf{R}_{\text{cm}}$ and, through an anisotropic polarizability, on orientation. Permutations of like nuclei remain exact symmetries; the actions of $SO(3)$ and E^* are still defined but commute with the Hamiltonian only as far as the trap allows; and photon recoil enters through $e^{i\mathbf{k} \cdot \mathbf{R}_{\text{cm}}}$.

Definition 3 (States). Let $\mathcal{H}_{\text{spin}} = \bigotimes_i \mathbb{C}^{2I_i+1}$, on which S acts by σ , permuting factors, with E^* trivial. Let $\varepsilon(g)$ be the sign of the permutation restricted to the fermionic nuclei, and let $\chi_+ = 1$ and $\chi_- = \det$ be the two *parity characters*. A *state* of parity $\pm$ is a square-integrable $\Psi : \mathcal{C} \rightarrow \mathcal{H}_{\text{spin}}$ obeying the *Pauli rule*²

$$\Psi(g \cdot X) = \varepsilon(g) \chi_\pm(g) \sigma(g) \Psi(X) \quad \text{for all } g \in S. \quad (1)$$

The Hamiltonian H is the kinetic energy in the mass metric plus a Born–Oppenheimer potential. We assume the potential real and invariant under $SO(3)$ and S , with a self-adjoint realization of H on

$$\mathcal{H} = L^2(\mathcal{C}) \otimes \mathcal{H}_{\text{spin}};$$

³ then H commutes with $SO(3)$ and S .

For unstarred g , (1) says that relabeled configurations are one physical state. For E^* it fixes the parity.

Fixed configurations. If $g \cdot X = X$, then (1) becomes a condition on the value $\Psi(X)$ (eigenfunctions are assumed continuous away from coincidences),

$$\sigma(g) \Psi(X) = \varepsilon(g) \chi_\pm(g) \Psi(X), \quad (2)$$

the *node rule*.⁴ Water’s straight configuration is fixed by $(23)^*$: at parity + only the proton singlet survives there (Figure 3).

²The symmetrization postulate, in permutation-inversion form.

³For example a potential bounded below and locally square-integrable, or Coulomb-like at coincidences [10].

⁴Known in few-body physics as inherent nodal surfaces.

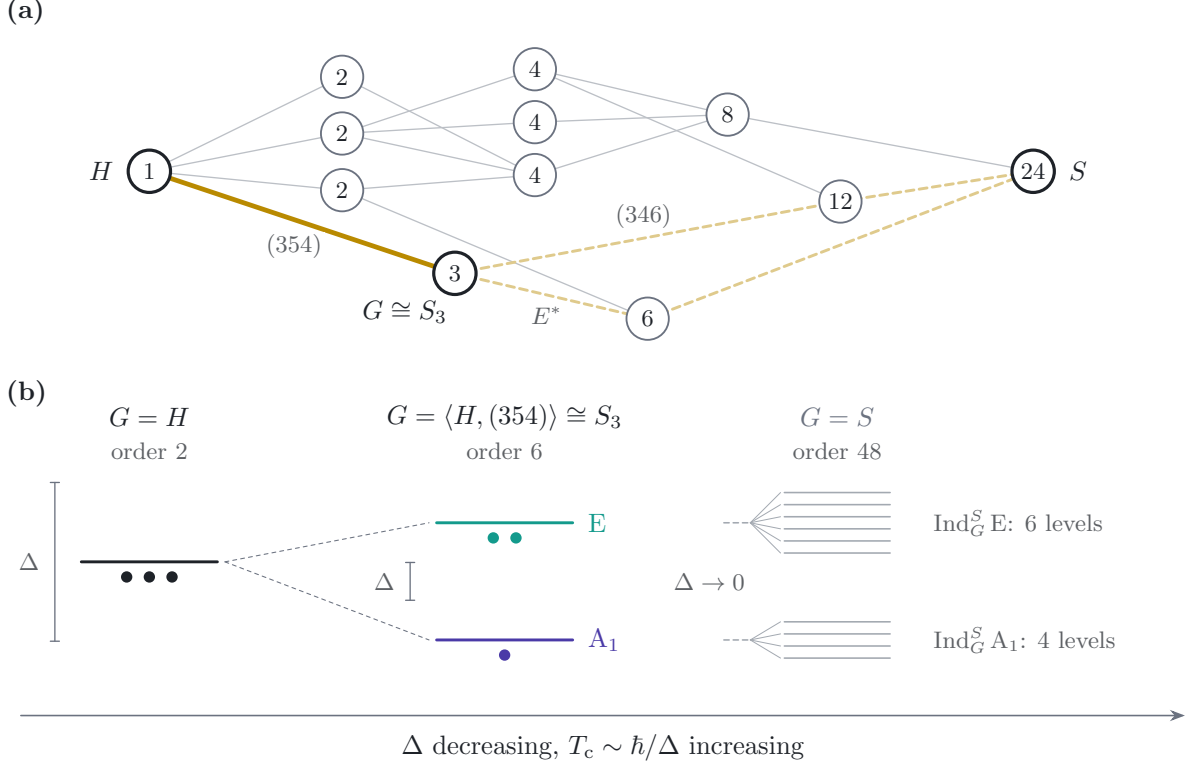


Figure 4: (a) Every group G' with $H \leq G' \leq S$ for methanol: 14 groups, drawn as 12 classes up to relabelings that fix H . The number in each node is $|G'/H|$, the versions, and a line joins classes whose representatives are contained one in the other (inclusions, not paths). Each highlighted edge adds one motion: the methyl turn (354) from H (solid), then E^* or (346), which moves H6, from S_3 (dashed), routes that would need splittings methanol does not show. (b) Methanol's ground torsional cluster at $J = 0$, three states (dots), at three resolutions Δ (bars, and $\Delta \rightarrow 0$ at right). Coarse, the states merge and $G = H$; resolving the splitting separates A_1 from E and picks $G \cong S_3$; resolving every splitting would pick all of S .

2 The regime

Definition 4 (Reference and point group). A *reference* is a configuration $X_0 \in \mathcal{C}$ with no two nuclei coincident. Its *point group* (Figure 3b), the stabilizer of the orbit $SO(3)X_0$, is

$$H = \{h \in S : h \cdot X_0 \in SO(3)X_0\},$$

found from X_0 alone. For $h \in H$ the rotation R_h with $h \cdot X_0 = R_h X_0$ is

$$R_h = (h \cdot X_0) \text{diag}(\mathbf{m}) X_0^\top \Lambda^{-1}, \quad \Lambda = X_0 \text{diag}(\mathbf{m}) X_0^\top, \quad (3)$$

with no fitting; for planar X_0 , Λ is inverted on the plane and the normal $\xi_1 \times \xi_2$ of an orthonormal pair ξ_1, ξ_2 in it goes to $R_h \xi_1 \times R_h \xi_2$, and for linear X_0 , R_h is fixed only up to a turn about the axis. The angle θ_h of R_h follows from $\text{Tr } R_h = 1 + 2 \cos \theta_h$.

Definition 5 (Motions and regime). A *motion* is a path from X_0 to a configuration $r(s \cdot X_0)$ with $r \in SO(3)$ and $s \in S$. Aligning the endpoint with X_0 by a rotation identifies s among the relabelings of like nuclei, up to H on either side, so a motion names a double coset HsH . A *regime* is a pair (X_0, G) with

$$G = \langle H, s_1, s_2, \dots \rangle,$$

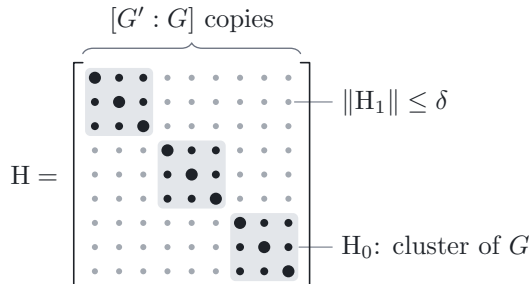


Figure 5: The effective Hamiltonian on the versions: the tinted blocks are H_0 , one copy of the cluster of G for each coset in G'/G , and the grey dots are the couplings H_1 left out.

generated by H and the motions that are resolved (Definition 6). The cosets G/H are the *versions*, the labeled copies of X_0 , up to rotation, that the molecule moves among.⁵

The candidates. Every regime lies in the interval $[H, S]$ of groups between H and S , and resolving one more motion moves up, $G \mapsto \langle G, s \rangle$. Each step adds double cosets, the tunneling parameters of Section 3, and its effect is exact: new versions, and each level of species Γ becoming levels carrying $\text{Ind}_G^{(G,s)} \Gamma$. Two kinds of motion follow from the bonds alone. Turning the fragment on one side of a bond whose removal disconnects the molecule, by an angle that maps the fragment onto itself, ends at a relabeled copy that is matched exactly; and the relabelings that preserve every bond bound all motions that break none. For methanol $H = \{E, (45)^*\}$, the methyl turn by a third matches $s = (354)$, and $\langle H, s \rangle \cong S_3$; above it the interval branches (Figure 4a). Adding E^* splits only E , into E' and E'' with spin weights $4/0$ and $0/4$, while A_1 and A_2 keep one level each. For eleven molecules the textbook group lies between these two bounds, at an end for ten:

molecule	$ H $	turns	$ G $	bonds	$ S $	between
water	4	4	4	4	4	1
ammonia	6	6	12	12	12	2
methanol	2	6	6	12	48	2
dimethyl ether	4	36	36	144	2880	3
methylamine	2	6	12	24	240	5
hydrazine	2	2	16	16	96	8
ethane	12	36	36	144	2880	3
acetone	4	36	36	144	8640	3
PF_5	12	12	240	240	240	4
methane	24	24	24	48	48	2
CH_5^+	2	2	240	240	240	48

with “turns” the order of H closed under the turns, “bonds” that of the bond-preserving relabelings, and “between” the number of groups from the one to the other. Methylamine’s group differs from the two other candidates of its order by how the amino inversion is combined with the torsion. This benchmarks the generator family on these examples; it does not show that an ordering is realized or unique. Which motions are resolved remains the physical input. Proposition 1 certifies a candidate’s dynamical accuracy by bounding what it leaves out; the outputs of Section 3 test it against data and tell candidates apart. Neither alone yields a unique group. Subgroups of S and their enlargement

⁵ G is the molecular symmetry group, Longuet-Higgins’s group of feasible relabelings [8]; “versions” and the “equivalent rotation” R_h are Bunker and Jensen’s terms [9]; Hougen’s “full point group” multiplies each point-group operation by such a rotation [11]. Hougen [12] builds G from generators of large-amplitude motions.

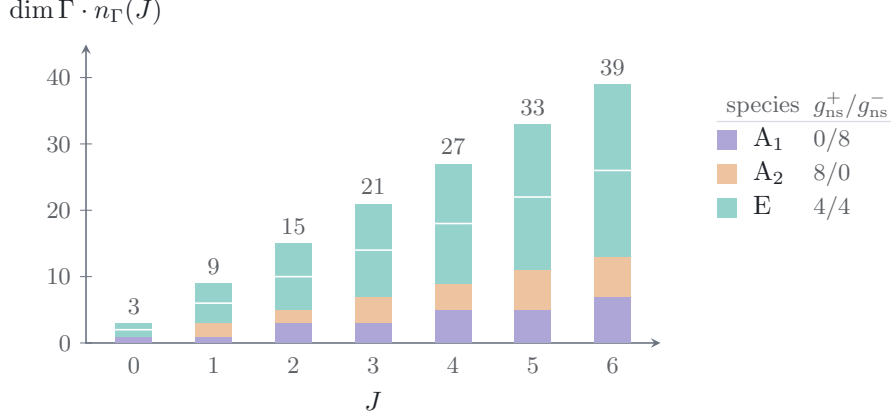


Figure 6: Methanol's levels at angular momentum J , stacked by species as $\dim \Gamma \cdot n_\Gamma(J)$; the totals are $|G/H|(2J+1) = 3(2J+1)$. Each E bar is halved by a white rule, one half per component. The key gives the spin weights g_{ns}^+/g_{ns}^- , the spin states a level of each species carries at parity + and -.

as tunneling is resolved are standard [8, 9, 13, 14]; added here is the interval as an explicit, checked search space whose nodes carry their versions, their fingerprints and, with Proposition 1, a bound on what they leave out, and whose action on $\mathcal{C}$ Part II rewrites in chart coordinates (6).

Definition 6 (Group from resolution). A *cluster* is the span of the eigenstates of H in an energy window with a gap on both sides. At resolution Δ , join neighbouring levels whose gap is less than Δ ; the chains formed this way are the merged multiplets. A feasible group is an inclusion-minimal G in $[H, S]$ for which every merged multiplet carries the induced representation $\text{Ind}_G^S \Gamma$ of an irreducible Γ . Its irreducible representations are the *species*.

S qualifies when the merged multiplets are irreducible. A proper G qualifies when the couplings outside it obey Proposition 1 (below) with $G' = S$ and $2\delta < \Delta$, and the levels of H_0 , each of one species, lie more than $\Delta + 2\delta$ apart. Several incomparable groups may qualify; for methanol `checks/build.py` shows that nonconjugate candidates in $[H, S]$ are told apart by their level counts and spin weights. Resolution is time: a splitting is resolved over a coherence time T_c when it exceeds about $\hbar/T_c$.⁶

Proposition 1 (Leaving couplings out). *Let $H \leq G \leq G'$, let a cluster's effective Hamiltonian on the cosets G'/H (one state each) be G' -equivariant, and write it as $H = H_0 + H_1$ (Figure 5), with H_0 the terms whose double cosets HgH lie in G and $\|H_1\| \leq \delta$. Then H_0 is $[G' : G]$ copies of the cluster of G ; every level of H lies within δ of one of H_0 ; a state starting in one copy leaves it by time t with probability at most $(\delta t/\hbar)^2$; and a level of species Γ isolated by more than 2δ becomes levels carrying $\text{Ind}_G^{G'} \Gamma$ (proof in Appendix A).*

Leakage below γ over time T_c is guaranteed if

$$\delta \leq \hbar\sqrt{\gamma}/T_c,$$

and for every normalized state the two evolutions stay close,

$$\|e^{-iHt/\hbar}\psi - e^{-iH_0t/\hbar}\psi\| \leq \delta t/\hbar,$$

so the expectation of a bounded observable changes by at most twice its norm times $\min(1, \delta t/\hbar)$; unbounded observables need further control of the state.

⁶Feasibility tied to the time scale of observation: Longuet-Higgins [8]; Wales and Salamon [15]; for the history of the term, and Watson's reading of unfeasible elements as uninformative, Bunker [11]. The spectral definition and the bound below are this work's.

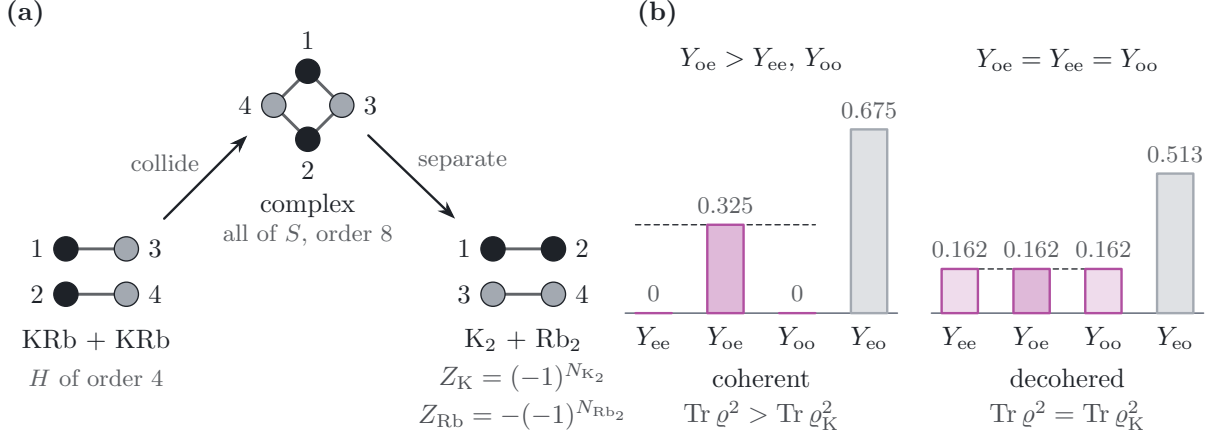


Figure 7: (a) Two KRb collide into a complex where every relabeling in S is feasible; the products K_2 and Rb_2 read the signs Z_K and Z_{Rb} . (b) Product yields Y_{xy} (e even, o odd) for an illustrative coherent field-mixed spin state and for the decohered mixture with the same populations. Only the coherent state lifts Y_{oe} above Y_{ee} (and Y_{oo}), certifying K–Rb spin entanglement. It shows a symmetry-compatible readout of spin information.

3 Outputs

All characters are sums of roots of unity, so every discrete output of the model is exact arithmetic in the cyclotomic integers.

Level counts. In the version model,⁷ where the cluster at angular momentum J is one localized vibrational ground state per version, the cluster carries $\text{Ind}_H^G D^J$ (Figure 6),⁸ and for nonlinear X_0 , by Frobenius reciprocity

$$n_\Gamma(J) = \frac{1}{|H|} \sum_{h \in H} \chi_\Gamma(h^{-1}) \chi_J(R_h), \quad \chi_J(R) = \frac{\sin((J + \frac{1}{2})\theta)}{\sin(\theta/2)}.$$

Spin weights. Species Γ at parity $\pm$ pairs with

$$g_{\text{ns}}^\pm(\Gamma) = \frac{1}{|G|} \sum_{g \in G} \chi_\Gamma(g) \chi_\sigma(g) \varepsilon(g) \chi_\pm(g), \quad \chi_\sigma(g) = \prod_{\text{cycles}} (2I + 1)$$

spin states, I the spin of the nuclei in each cycle of g . Summed with the counts,

$$\sum_\Gamma n_\Gamma(J) (g_{\text{ns}}^+ + g_{\text{ns}}^-) = \frac{1}{|H|} \sum_{h \in H} \chi_J(R_h) \chi_\sigma(h) \varepsilon(h) (1 + \det h),$$

whose leading term at large J , when H has a starred element, is $(2J + 1) \dim \mathcal{H}_{\text{spin}} / |H^+|$, with H^+ the unstarred elements of H .

Selection rules. The dipole carries $\det$; a line joins Γ and Γ' only if $\Gamma' = \Gamma \otimes \det$, so lines never change the nuclear-spin species (ortho stays ortho).

⁷The tunneling-matrix formalism of Hougen and coworkers [5]: couplings organized by double cosets HgH .

⁸So the unsplit cluster already carries a representation of the larger group, and resolving a splitting only reduces it; this answers the apparent paradox that splitting comes with a larger group [14].

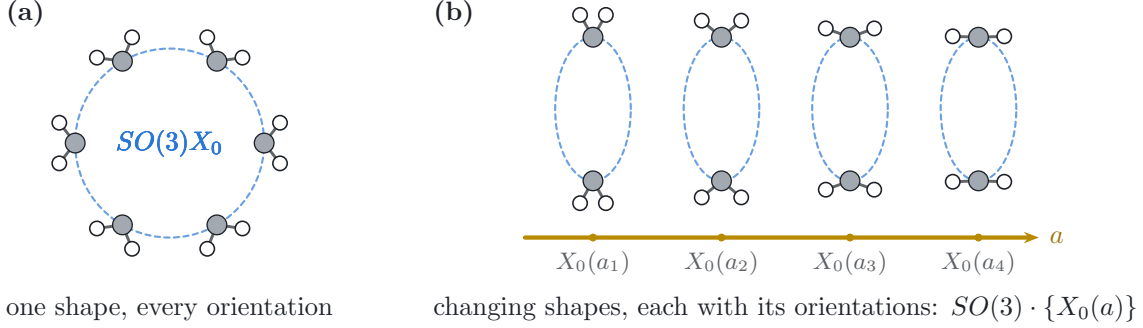


Figure 8: Rigid and nonrigid. (a) The rigid-orientation setting: one shape and its orbit $SO(3)X_0$ under rotations. (b) Here: a family of shapes $X_0(a)$, each with its own orbit of orientations; the symmetry acts on the whole family, so orientation and internal coordinates are coupled rather than an independent product.

Several molecules. Nothing above assumed one molecule. For separated fragments H is the set of relabelings mapping fragments onto fragments as rotated copies, and a reaction moves through the subgroups of S . For $\text{KRb} + \text{KRb} \rightarrow \text{K}_2 + \text{Rb}_2$, with nuclear spins as spectators⁹ and equal yields from each spin sector (an added assumption), take two independent copies $\varrho \otimes \varrho$ of the nuclear-spin state ϱ of one KRb, with reduced states ϱ_{K} and ϱ_{Rb} . With N_{K_2} and N_{Rb_2} the products’ rotational quantum numbers, the signs from the Pauli rule for each product,

$$Z_{\text{K}} = (-1)^{N_{\text{K}_2}}, \quad Z_{\text{Rb}} = -(-1)^{N_{\text{Rb}_2}},$$

have the joint law

$$\Pr(Z_{\text{K}}, Z_{\text{Rb}}) = \frac{1}{4} (1 + Z_{\text{K}} \text{Tr } \varrho_{\text{K}}^2 + Z_{\text{Rb}} \text{Tr } \varrho_{\text{Rb}}^2 + Z_{\text{K}} Z_{\text{Rb}} \text{Tr } \varrho^2).$$

A separable ϱ has $\text{Tr } \varrho^2 \leq \text{Tr } \varrho_{\text{K}}^2, \text{Tr } \varrho_{\text{Rb}}^2$, so $Y_{\text{oe}} > Y_{\text{ee}}$ or $Y_{\text{oe}} > Y_{\text{oo}}$, with Y_{xy} the fraction of products with N_{K_2} of parity x and N_{Rb_2} of parity y , certifies K–Rb spin entanglement inside one KRb (Figure 7).¹⁰

Against published data. Spin weights need no fitted constant, so they can fail. Computed from the list of nuclei and (X_0, G) alone, they reproduce the published values [18, 9]:¹¹

molecule	spin weights of the species
water	para 1, ortho 3
ammonia (¹⁴ N)	A ₁ 0, A ₂ 12, E 6
acetylene	para 1, ortho 3
methane	A 5, E 2, F 3
benzene	A ₁ 7, A ₂ 3, B ₁ 13, B ₂ 1, E ₁ 11, E ₂ 9

Inputs, in one place.

1. The list of nuclei: types (mass, charge, spin). This fixes $\mathcal{C}$, S , $\mathcal{H}_{\text{spin}}$ and the Pauli rule.
2. A reference X_0 and the motions resolved by experiment. This fixes H , G , the versions, the version-model level counts and the spin weights.

⁹As observed by Hu et al. [16]. Figure 7 uses an illustrative state of the field-mixed form they prepare; its numbers are not taken from that experiment.

¹⁰The purity, or Rényi-2 entropy, criterion for separability [17], read out as a swap test through the reaction.

¹¹The comparison is the script `checks/data_tests.py`.

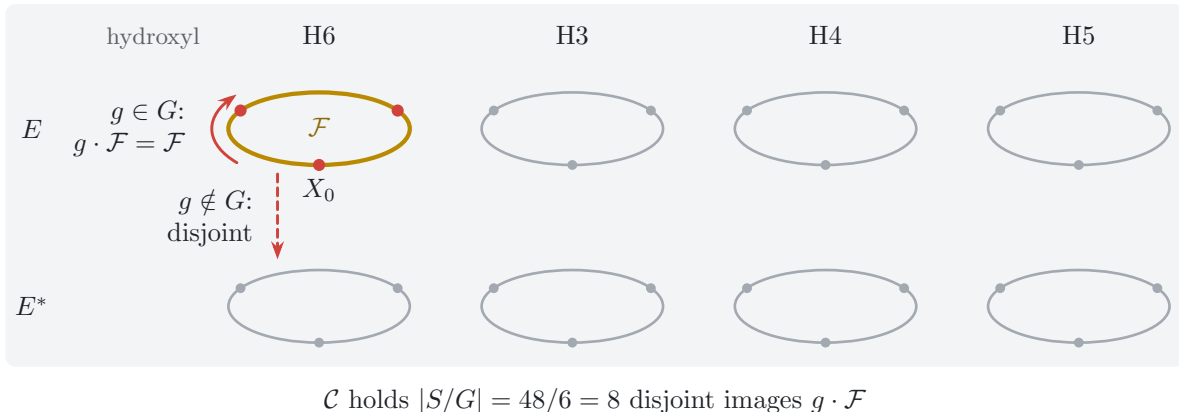


Figure 9: Methanol’s $\mathcal{F}$ (amber) and its images $g \cdot \mathcal{F}$ (grey), one for each coset in S/G , each drawn as its circle of torsion angles with three versions. Columns say which hydrogen sits at the hydroxyl site, and the lower row holds the mirror images.

3. The route of each motion’s path. With the masses this fixes the version graph, $\mathcal{F}$, $\mathcal{P}$, $\hat{\mathcal{P}}$ and the kinetic constants (ρ , holonomy; Part II).
4. A potential or spectral data. This fixes the energies, for methanol V_3 and the band.

Choices that change nothing measurable: the generators of G with their paths carried along, the frame along the paths, and the labels.

What is exact. The group actions and the Pauli rule are exact in the stated setting; counts and spin weights are exact arithmetic within the version model; the charts of Part II are exact on their domains. Band energies come from an effective model, the reaction readout assumes the stated preparation and equal sector yields, and the data comparison is reproducible from the scripts.¹²

Part II. Coordinates

States, symmetry constraints and the Hamiltonian are defined on the ambient configuration space, so Part I needs no adapted coordinates. Adapted charts resolve orientation, internal motion and normal deformation for calculation (frame turns, bands) without changing that underlying physical description; this part builds them, and the symmetry acts on them by explicit formulas.

The rigid limit. With internal shape frozen, $\mathcal{A}$ is a point, $\mathcal{F} = SO(3)X_0G$, and restricting to the rotational orbit $SO(3)X_0$ with the same spin-statistics constraints recovers the setting of molecular orientation theory (Figure 8a). Nonrigid motion enlarges the configuration domain and couples orientation to internal coordinates (Figure 8b); it does not merely add an independent tensor factor. Two geometric effects should be kept apart: the symmetry-induced monodromy of position states on a rigid orbit, studied there, and the holonomy of the mechanical connection along deforming shapes, met here in Section 8. The second is not shown here to give a protected operation.

¹²Nuclear types and input geometries (`checks/molecules.py`), group generators and character conventions (`checks/groups.py`, `checks/build.py`) and the full check output (`checks/checks.txt`) are in the source of this paper. Exact quantities use rational arithmetic; continuous ones use the stated geometries in double precision, with tolerances printed beside each check. Displayed matrices are rounded.

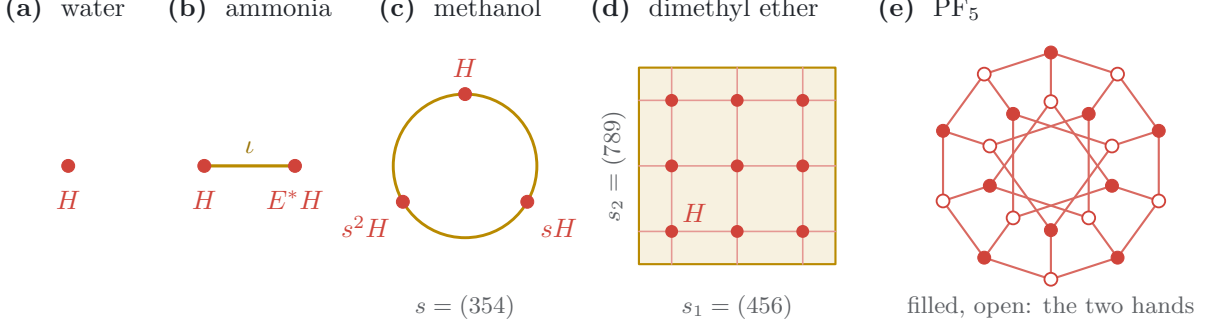


Figure 10: Version graphs: one vertex per version gH , joined to the versions in $gHsH$ for each motion s . (a) Water, one version. (b) Ammonia, two, joined by an inversion. (c) Methanol, three on a circle. (d) Dimethyl ether, nine on the torus of two methyl turns. (e) PF₅, twenty on the Desargues graph, where every Berry step changes hand. PF₅ shows a collective motion organized by its pathways rather than by independent rotor circles.

4 The feasible configuration space

Closing motions and inversions. A motion *closes* if its path, repeated without retracing, returns to a rotated copy of X_0 after f steps (a torsion or a pseudorotation of order f). Then $X_0(\tau + 2\pi)$ is a rotated copy of $X_0(\tau)$, and s acts as $\tau \mapsto \tau + 2\pi/f$. It is an *inversion* if $s^2 \in H$ and the path passes a planar arrangement; its path is then an interval, $X_0(\iota)$ with $\iota \in [-1, 1]$, which s reflects. For methanol, $s = (354)$ turns the methyl by a third, so $f = 3$; ammonia’s umbrella is an inversion.

Definition 7 (Feasible configuration space). The *internal coordinates* a of the chosen motions range over a *reference domain* $\mathcal{A}$, and $X_0(a)$ is the *reference family*; for n_τ independent closing motions and n_ι inversions, $a = (\tau, \iota)$ and

$$\mathcal{A} = \mathbb{T}^{n_\tau} \times [-1, 1]^{n_\iota}, \quad \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}.$$

Then

$$\mathcal{F} = SO(3) \cdot \{X_0(a)\} \cdot G \subset \mathcal{C},$$

with the reference family assumed nonsingular. $\mathcal{F}$ is invariant under G by construction, and we require it to be *feasible* for G :

$$g \cdot \mathcal{F} \cap \mathcal{F} = \emptyset \text{ for } g \notin G.$$

This compatibility of the chosen region with the spectral regime is a hypothesis to check; it does not follow from Definition 6. When $\mathcal{A}$ is a point, $\mathcal{F} = SO(3)X_0G$ is $|G/H|$ disjoint copies of $SO(3)$ ($SO(3)/SO(2)$ for linear X_0); for a rigid molecule $G = H$ and it is one.

The images $g \cdot \mathcal{F}$, one for each coset in S/G , are the disjoint regions of $\mathcal{C}$ that the molecule could occupy in this regime (Figure 9).

Two checks on the chosen motions: each is independent of the others (two methyls turn separately), and when they commute and their steps give distinct cosets exhausting G/H , they count the versions:

$$|G/H| = \left(\prod_{\text{circles}} f \right) \cdot 2^{\#\text{inversions}}.$$

The *version graph* joins each version gH to the versions in $gHsH$, for each motion s (Figures 10 and 11). G acts on it, and its edges are the leading tunneling couplings of Section 3, one double

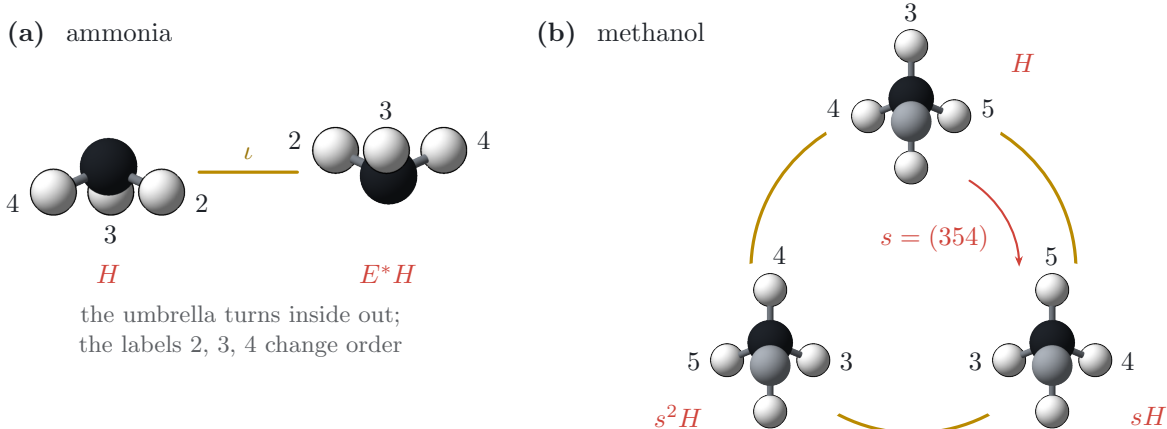


Figure 11: Versions drawn as molecules. Each is $g \cdot X_0$: the same points as X_0 with its labels permuted by g , inverted through the centre for E^* . (a) Ammonia’s two versions, joined by the umbrella coordinate ι . (b) Methanol’s three, on the circle of the methyl turn $s = (354)$. Ammonia shows a nonperiodic inversion path, on which s acts by the reflection $\iota \mapsto -\iota$.

coset HsH per chosen motion. When the motions commute it is a grid on the torus of the circles, times one edge for each inversion. PF_5 ’s Berry steps do not commute, and its graph is the Desargues graph. Another generating set traces the same $\mathcal{F}$ in coordinates changed by an integer matrix; what is physical is the route of each path, which its endpoints alone do not fix.

5 Position space

$\mathcal{C}$ is flat, and the matrix X is a global coordinate on it in which both actions are linear. Position space is a second coordinate system, adapted to orientation and motion. A nonidentity rotation fixes a configuration exactly when $\text{rank } X \leq 1$, a linear arrangement. Linear configurations are regular points of $\mathcal{C}$; their axial stabilizer makes an independent orientation and shape redundant, and the associated-space chart below handles that redundancy without a singular Hamiltonian. The stabilizer and the bending angular momentum it brings are genuine structure. Other failures belong to the chart, not to $\mathcal{C}$: we choose a reference family that is a smooth embedded patch, and restrict Q below so that its tube is parametrized one-to-one. If the molecule reaches a linear arrangement, X_0 is taken there, the turns about its axis are carried as a coordinate (below), and the bent configurations near it are ordinary points of the chart.¹³

Normal coordinates. At each point of the reference family the *normal directions* are those mass-orthogonal both to its rotational orbit and to the tangents of the internal motions; on a regular nonlinear patch they have dimension $q = 3N - 6 - \dim \mathcal{A}$ ($3N - 5 - \dim \mathcal{A}$ for a linear X_0). The reference family is specified only up to a rotation at each internal point a , and that choice is the *body frame*. The columns of $L(a)$ are a chosen orthonormal basis of the normal directions, varying continuously with a , and $Q = (Q_1, \dots, Q_q)$ are the *normal coordinates*; L is part of that choice. $\mathcal{A}$ parametrizes the reference shapes; their body-frame representatives $X_0(a)$ are defined on patches, or on an unrolled interval of τ , and are matched by the transition rotations in (8) below. The *reference point* a_* is the one with $X_0(a_*) = X_0$: for methanol $a_* = 0$; for ammonia $a_* = 1$, its other version sits at $\iota = -1$, and the midpoint $\iota = 0$ is the planar inversion configuration.

¹³This is the slice theorem [19] for $SO(3) \times G$ acting on $\mathcal{C}$. For a linear X_0 it gives the bender of Hougen, Bunker and Johns [2], with its condition $K = \ell$.

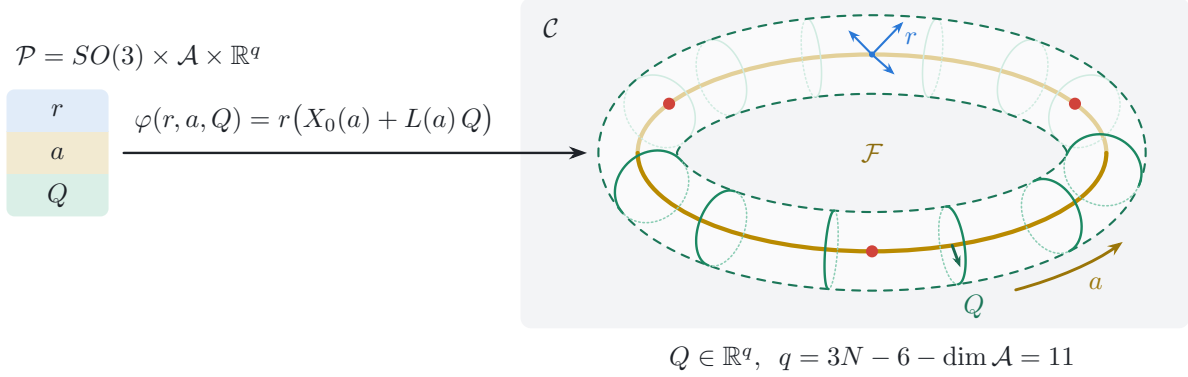


Figure 12: The chart φ maps $\mathcal{P}$ onto a tube (white, dashed outline, drawn see-through) around $\mathcal{F}$. The gold circle stands for the reference shapes $X_0(a)$, with methanol's three versions (red); the blue axes mark one orientation r ; each ring is a cross-section, the normal coordinates Q ($q = 11$ of them, drawn as two).

Definition 8 (Position space). The *position space* is the local coordinate model

$$\mathcal{P} = SO(3) \times \mathcal{A} \times \mathbb{R}^q,$$

with Q restricted to a neighbourhood of 0 on which the tube is one-to-one, mapped into $\mathcal{C}$ by the chart (Figure 12)

$$\varphi(r, a, Q) = r(X_0(a) + L(a)Q). \quad (4)$$

For methanol $\mathcal{A} = \mathbb{T}$, for ammonia $\mathcal{A} = [-1, 1]$. The product names this local model; the transition maps below connect patches, and the images under G cover a neighbourhood of $\mathcal{F}$.¹⁴ Its points $|r, a, Q\rangle$ are the *position basis*. For a linear X_0 with $\mathcal{A}$ a point the turns $R_\theta \in SO(2)$ about its axis fix it, and $\mathcal{P} = SO(3) \times_{SO(2)} \mathbb{R}^q$ with $[rR_\theta, Q] = [r, R_\theta Q]$, where R_θ acts on Q through the normal frame, turning the degenerate bending pairs.

The frame. Along each closing motion we take the frame with zero total angular momentum. For a symmetric end group turning about the unit axis λ the frame turns back about the *frame vector*

$$\rho = I_\alpha I^{-1} \lambda \quad (5)$$

by $\rho = |\rho|$ per unit τ (Figure 13), with I the inertia tensor and I_α the end group's moment; for methanol $\rho = 0.810$.¹⁵ Over several torsions this frame may not close; the frame is then any smooth choice, and what remains is the curvature of the mechanical connection (Section 8, Figure 17).¹⁶

The action of G . Since $g \cdot (rX) = r(g \cdot X)$, g acts through the reference family. We take that family closed under G up to rotation,

$$g \cdot X_0(a) = R_g(a) X_0(g \cdot a)$$

(otherwise the same formula relates its G -translated patches), and

$$g : (r, a, Q) \mapsto (r R_g(a), g \cdot a, L_g(a) Q), \quad (6)$$

¹⁴This is the large-amplitude reference path of Hougen, Bunker and Johns [2], and for reactions the reaction-path Hamiltonian of Miller, Handy and Adams [3]; here it carries explicit symmetry actions.

¹⁵The vector ρ of the rho-axis method. The zero-angular-momentum frame is the internal-axis method. Both are reviewed by Lin and Swalen [4].

¹⁶The same obstruction led Hougen to local internal-axis frames, one near each framework and one along each tunneling path [5]; a purely vibrational motion can end in an arbitrary rotation [6].

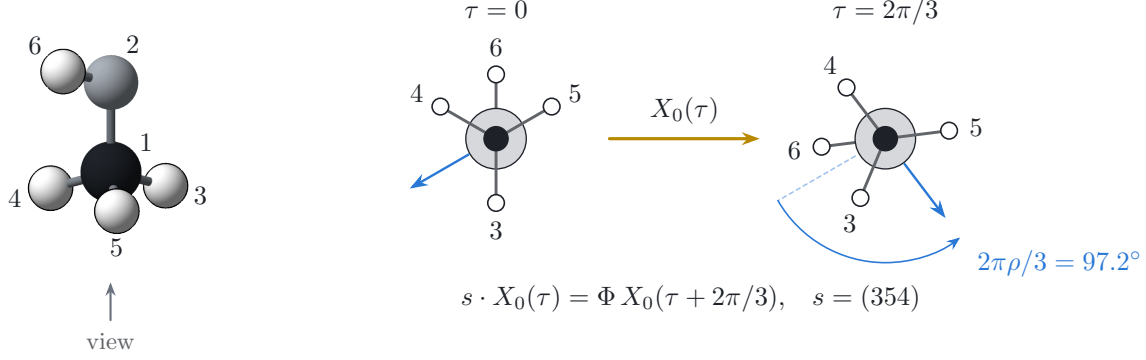


Figure 13: Methanol (left) and its torsion seen along the arrow, down C1 to O2, in the zero-angular-momentum frame. Along the path $X_0(\tau)$ the methyl and the rest of the molecule turn oppositely, by $2\pi/3$ relative to each other; the rest, with the axis fixed to it (blue), turns by $2\pi\rho/3$. So the relabeled reference $s \cdot X_0(\tau)$ is the later point of the path turned by Φ . It shows the explicit torsion action: s shifts τ by a third of a turn and turns the frame by the fixed Φ .

with $R_g(a)$ given by (3) (with $g \cdot X_0(a)$ for $h \cdot X_0$ and $X_0(g \cdot a)$ for X_0) and $L_g(a)$ the matrix of $R_g(a)^{-1}g$ between the normal frames at a and $g \cdot a$. For methanol $s \cdot \tau = \tau + 2\pi/3$; for ammonia $s \cdot \iota = -\iota$. The turns compose as

$$R_{gh}(a) = R_h(a) R_g(h \cdot a), \quad (gh) \cdot a = g \cdot (h \cdot a). \quad (7)$$

For $h \in H$, $R_h(a_*)$ is R_h of (3). For a symmetric-top torsion, $R_s = \Phi$, the turn by $2\pi\rho/f$ about ρ , and since $s^f = E$ the circle is glued with a turn,

$$(r, \tau + 2\pi, Q) \sim (r\Phi^{-f}, \tau, \Omega(\tau) Q), \quad L(\tau) \Omega(\tau) = \Phi^f L(\tau + 2\pi), \quad (8)$$

with $\Omega(\tau)$ the transition between the normal frames; in a frame carried along the circle it is the identity. The shape loop closes, and its representative closes up to rotation,

$$X_0(\tau + 2\pi) = \Phi^{-f} X_0(\tau).$$

By (7), (6) obeys every group relation of G once each circle is glued by (8).

States on the chart. A state is a function $\psi : \mathcal{P} \rightarrow \mathcal{H}_{\text{spin}}$ compatible with the gluing and obeying the Pauli rule in chart form,

$$\psi(g \cdot (r, a, Q)) = \varepsilon(g) \chi_{\pm}(g) \sigma(g) \psi(r, a, Q),$$

with $g \cdot (r, a, Q)$ given by (6). The versions are the $|G/H|$ images of a_* . Chart wavefunctions use the measure induced from the mass metric on $\mathcal{C}$,

$$d\mu_{\mathcal{C}} = |\det D\varphi| dr da dQ,$$

with dr the Haar measure and $D\varphi$ written in frames orthonormal for the mass metric and for the product metric on the chart; multiplying by $|\det D\varphi|^{1/2}$ moves them to the product measure, with the kinetic operator transformed alongside.

Fixed points. Since φ commutes with G , an element fixes a point of $\mathcal{P}$ only if it fixes its configuration. A symmetry realized by a rotation, $h \cdot X_0 = R_h X_0$ with $R_h \neq 1$, acts on orientation by $r \mapsto r R_h$, which fixes no r . What remains are configurations with two like nuclei coincident, or mapped onto themselves by a starred element, as at an inversion centre. They form a set of measure zero and enter only through the node rule (2).

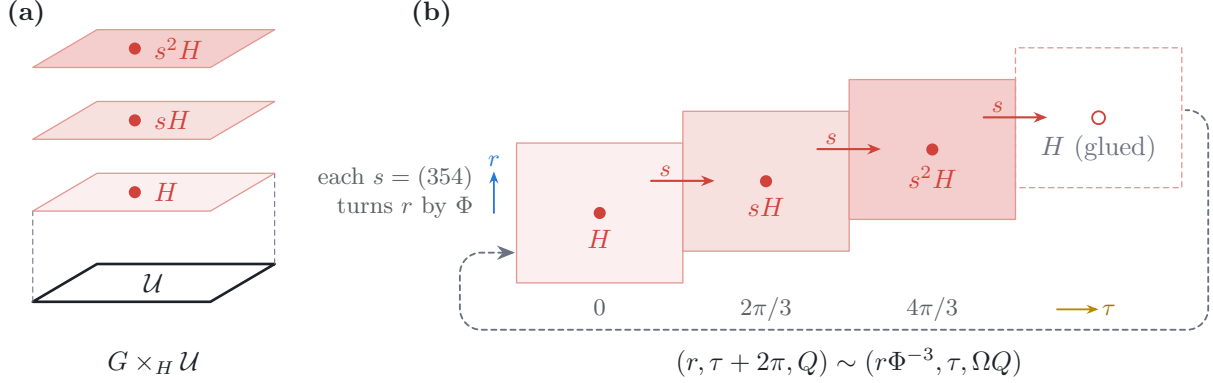


Figure 14: (a) The deck stacked: one copy of the cell $\mathcal{U}$ for each version gH , which is the induced space $G \times_H \mathcal{U}$. (b) The same cards laid out along τ : neighbours meet on walls where the coordinates change by the action of s , each step turning the frame by Φ , and the gluing closes the circle at the seam, re-expressing the normal coordinates by Ω .

The atlas. When the region admits an H -invariant cell whose G/H translates partition it up to boundaries, the cells form a deck, as for methanol; otherwise ordinary overlapping patches serve. Let $\mathcal{U} \subset \mathcal{P}$ be such a cell: every orientation, small Q , and the internal coordinates in a cell of $\mathcal{A}$ around a_* (for a torsion, within half a step). H maps $\mathcal{U}$ to itself, and $g \cdot \mathcal{U}$ depends only on the coset gH . The charts therefore form a deck indexed by the versions, the coset space G/H , and G permutes the cards by left multiplication (Figure 14). The deck can be read two ways. Laid out, the cards $g \cdot \mathcal{U}$ meet their neighbours on walls, where the change of coordinates is the action (6), with the gluing (8) at the seam. Stacked, the molecule's region is

$$G \times_H \mathcal{U} = \{[g, u]\}, \quad [gh, u] = [g, h \cdot u], \quad [g, u] \mapsto g \cdot \varphi(u), \quad (9)$$

one copy of $\mathcal{U}$ per version. The bracket only says that two names denote one configuration. The induced space alone does not join neighbouring cards: the connected region is $(G \times_H \mathcal{U})/\sim$, with each wall of a card identified with its neighbour's through (6) and (8).¹⁷ The walls have measure zero, so square-integrable functions on the region are the induced representation Ind_H^G of functions on $\mathcal{U}$; the Hamiltonian's domain adds the matching of wavefunctions and of their normal flux across the walls. Keeping one low state per card is the version model of Section 3.

6 Momentum space

Definition 9 (Momentum space). For one torsion, in the vibrational ground sector ($\mathbf{v} = 0$), expanding in Wigner functions on $SO(3)$ and in waves along the circle gives the *momentum space*¹⁸

$$\hat{\mathcal{P}} = \{(J, M, K, \kappa) : \kappa \in \mathbb{Z} + \rho K\},$$

the labels of Wigner functions $D_{MK}^J(r)$ on $SO(3)$ ($J \in \mathbb{N}_0$, $-J \leq M, K \leq J$), of waves $e^{i\kappa\tau}$ along the circle of one torsion, whose wavenumbers the gluing shifts by ρK (below). These label a basis,

¹⁷ $G \times_H \mathcal{U}$ is the induced G -space, the geometric form of induction. When no configuration in the region is fixed by an element other than E , G also acts by deck transformations of the region over its quotient: a loop in the quotient lifts to a path from X to $g \cdot X$. For $N \geq 3$ the coincident and fixed configurations have codimension at least 3, so the rest of $\mathcal{C}$ is simply connected, and on it S is the full fundamental group of the quotient.

¹⁸Peter-Weyl on $SO(3)$ and Fourier series on the circle; excited vibrations add vibrational quantum numbers $\mathbf{v} \in \mathbb{N}_0^q$.

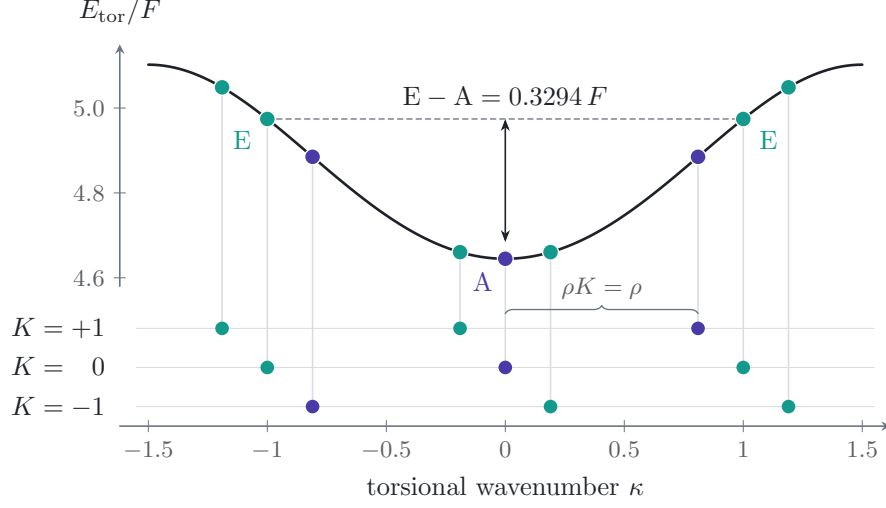


Figure 15: Methanol's torsional band $E_{\text{tor}}(\kappa)$ over one period, read at the shifted lattice $\kappa \in \mathbb{Z} + \rho K$ drawn below it for $K = +1, 0, -1$; each row is moved by ρK . Colour gives the species from $k = \kappa - \rho K$ modulo 3: A for $k \equiv 0$, E for $k \equiv \pm 1$. The arrow is the splitting $E - A$ at $K = 0$.

the *momentum basis* $|JMK, \kappa\rangle$, on which G acts by matrices. Several torsions or excited vibrations change the boundary condition, and with it the shift.

A point-group element acts on $\hat{\mathcal{P}}$ by

$$|JMK\rangle \mapsto \sum_{K'} D_{K'K}^J(R_h) |JMK'\rangle,$$

by $\tau \mapsto h \cdot \tau$ on the waves, and by L_h on the vibrational quantum numbers $\mathbf{v}$. M , fixed in the laboratory, is untouched. For a linear X_0 the identification $[rR_\theta, Q] = [r, R_\theta Q]$ keeps only $K = \ell$, the angular momentum of the bending pairs.

The torsion operator. Conventions: the body axis z' lies along ρ ; Φ turns by $2\pi\rho/f$, so $D_{K'K}^J(\Phi) = \delta_{K'K} e^{-2\pi i K \rho/f}$; a state is $\sum_K D_{MK}^J(r) \psi_K(\tau)$; and the operation s acts on it by pulling back (6),

$$(s\psi)(r, \tau) = \psi(r\Phi, \tau + 2\pi/f).$$

In the vibrational ground sector, (8) gives the twisted boundary condition

$$\psi_K(\tau + 2\pi) = e^{2\pi i \rho K} \psi_K(\tau), \quad \kappa \in \mathbb{Z} + \rho K,$$

and

$$s e^{i\kappa\tau} |JMK\rangle = \zeta_f^k e^{i\kappa\tau} |JMK\rangle, \quad k = \kappa - \rho K \in \mathbb{Z}, \quad \zeta_f = e^{2\pi i/f}.$$

So s^f acts as 1, as the relation $s^f = E$ requires, with no condition on Φ^f , and the circle labels of $\hat{\mathcal{P}}$ form the shifted lattice $\mathbb{Z} + \rho K$ (Figure 15). The quasi-momentum $k \pmod f$ labels the species of the cyclic group generated by s : for methanol A₁ or A₂ at $k \equiv 0$, E at $k \equiv \pm 1$.

For methanol ($f = 3$) let $E_{\text{tor}}(\kappa)$ be the lowest eigenvalue of $-F\partial_\tau^2 + \frac{V_3}{2}(1 - \cos 3\tau)$ on functions with $\psi(\tau + 2\pi/3) = e^{2\pi i \kappa/3} \psi(\tau)$, where F is the torsional constant and V_3 the barrier.¹⁹ The leading levels are

$$E_{\text{rot}}(J, K) + E_{\text{tor}}(k + \rho K),$$

¹⁹Here $F = 27.6468 \text{ cm}^{-1}$ and $V_3 = 373.555 \text{ cm}^{-1}$, from the global fit of Xu et al. [20], whose $\rho = 0.81021$ agrees with ours; they set the scale and are not a test.

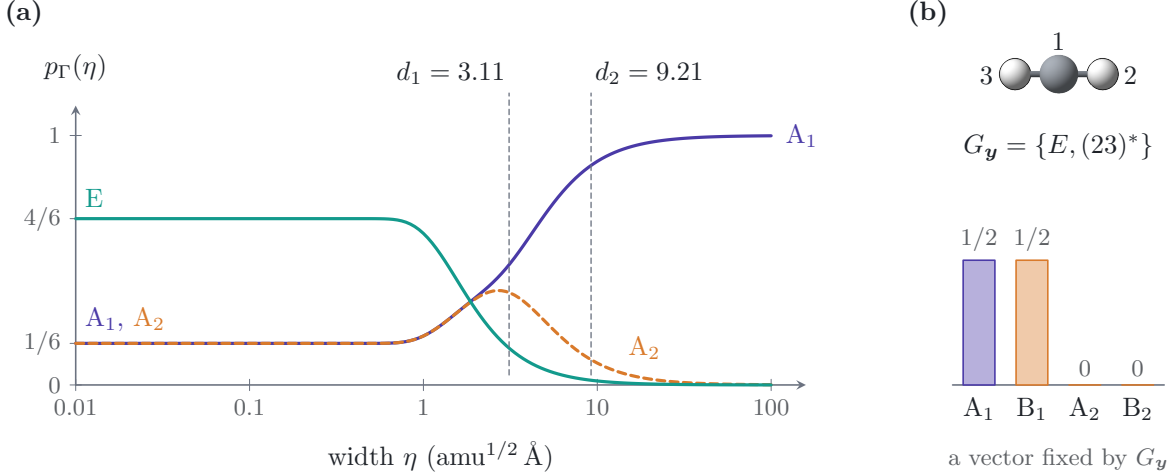


Figure 16: (a) Shares $p_\Gamma(\eta)$ of the species in a methanol wavepacket against its width: narrow packets hold the shares $(\dim \Gamma)^2/|G|$, and as the packet reaches its relabeled copies at d_1 and d_2 only A_1 remains. (b) At water's straight arrangement a narrow spatial packet holds only species with a vector fixed by the stabilizer $G_{\mathbf{y}}$: half A_1 , half B_1 .

with E_{rot} the symmetric-top energy (Figure 15). For two torsions the operators commute, as their relabelings do.

7 Coherent states

Position states are sharp in X and momentum states in $(J, K, \kappa, \mathbf{v})$. Gaussian packets are localized in the mass-weighted configuration and its conjugate momentum at once, and they are defined on all of $\mathcal{C}$ without a chart.

Definition 10 (Coherent states). In mass-weighted coordinates $\mathbf{y} = X \text{diag}(\mathbf{m})^{1/2}$ the kinetic metric on $\mathcal{C}$ is Euclidean, and rotations and relabelings act by orthogonal maps, since like nuclei have equal masses. The Gaussian states $|\mathbf{y}, \mathbf{p}\rangle \propto e^{-|\mathbf{y}' - \mathbf{y}|^2/2\eta^2 + i\mathbf{p} \cdot \mathbf{y}'/\hbar}$ of width η , as functions of $\mathbf{y}' \in \mathcal{C}$, are therefore defined on all of $\mathcal{C}$, and the symmetries permute them:

$$r|\mathbf{y}, \mathbf{p}\rangle = |r\mathbf{y}, r\mathbf{p}\rangle, \quad g|\mathbf{y}, \mathbf{p}\rangle = |g \cdot \mathbf{y}, g \cdot \mathbf{p}\rangle.$$

An admissible state built from $|\mathbf{y}, \mathbf{p}\rangle$ is its projection by the Pauli rule: one wavepacket at every relabeled copy $g \cdot \mathbf{y}$, with relative weights fixed by the species.

Proposition 2 (Species content of a wavepacket). *For $\mathbf{p} = 0$ the share of species Γ of G in $|\mathbf{y}, 0\rangle$ is*

$$p_\Gamma(\eta) = \frac{\dim \Gamma}{|G|} \sum_{g \in G} \chi_\Gamma(g^{-1}) e^{-|\mathbf{y} - g \cdot \mathbf{y}|^2/4\eta^2}.$$

The shares sum to 1. As $\eta \rightarrow 0$ only the stabilizer $G_{\mathbf{y}} = \{g \in G : g \cdot \mathbf{y} = \mathbf{y}\}$ survives,

$$p_\Gamma \rightarrow \frac{\dim \Gamma |G_{\mathbf{y}}|}{|G|} \dim \Gamma^{G_{\mathbf{y}}},$$

with $\Gamma^{G_{\mathbf{y}}}$ the vectors of Γ fixed by $G_{\mathbf{y}}$. For trivial $G_{\mathbf{y}}$ this is the Plancherel share $(\dim \Gamma)^2/|G|$. As $\eta \rightarrow \infty$ only the trivial species remains.

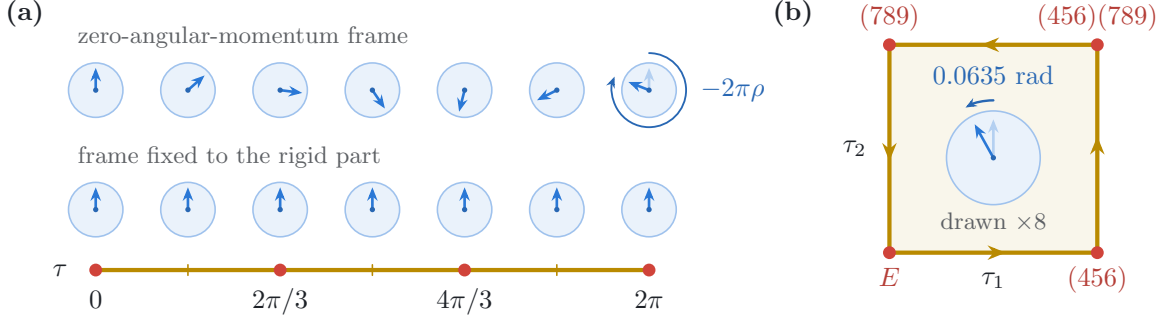


Figure 17: (a) Methanol’s circle unrolled; each needle is the body axis $\mathbf{x}'$ seen down the rotor axis. The zero-angular-momentum frame comes back turned by $-2\pi\rho$, so ρ sits in the gluing; a frame fixed to the rigid part closes, and ρ moves into a cross term. (b) In dimethyl ether the loop (456), (789), back, back turns the frame by 0.0635 rad, the holonomy of the mechanical connection, drawn eight times larger. Dimethyl ether shows commuting relabelings alongside curved mechanical transport: the group closes, the frame does not.

Proof. The projector onto species Γ is

$$P^\Gamma = \frac{\dim \Gamma}{|G|} \sum_g \chi_\Gamma(g^{-1}) g,$$

and two Gaussians of equal width overlap by $e^{-|\mathbf{y}-\mathbf{y}'|^2/4\eta^2}$. The limits follow from $\sum_{g \in G_{\mathbf{y}}} \chi_\Gamma(g^{-1}) = |G_{\mathbf{y}}| \dim \Gamma^{G_{\mathbf{y}}}$ and from $\chi_\Gamma(E) = \dim \Gamma$. $\square$

The finite-group projections of these packets give exact spatial species shares, including the narrow-packet Plancherel limit at a trivial stabilizer and, at a symmetric centre, the exclusion of every species without a vector fixed by the stabilizer. Physical packets additionally require nuclear-spin pairing and Pauli projection. Symmetry-related peaks under unstarred relabelings describe the same unlabeled nuclear arrangement, not separately addressable configurations. Gaussians in Cartesian coordinates [21] and symmetry-projected all-particle Gaussians [22] are established tools; what is added here is their species content under the regime group G and its stabilizers.

Methanol. For $\mathbf{y}$ at $X_0 \text{diag}(\mathbf{m})^{1/2}$, the copies $g \cdot \mathbf{y}$, $g \in G$, lie at distance $d_1 = 3.11$ (the two 3-cycles) and $d_2 = 9.21$ (the three starred transpositions), in $\text{amu}^{1/2} \text{\AA}$. With $c_1 = e^{-d_1^2/4\eta^2}$ and $c_2 = e^{-d_2^2/4\eta^2}$,

$$p_{A_1} = \frac{1}{6}(1 + 2c_1 + 3c_2), \quad p_{A_2} = \frac{1}{6}(1 + 2c_1 - 3c_2), \quad p_E = \frac{2}{3}(1 - c_1).$$

A packet with $\eta \leq d_1/4$ holds the species in proportion 1 : 1 : 4 to within $\frac{2}{3}e^{-4}$, and a wider one begins to see its copies and turns into A_1 (Figure 16). At a symmetric centre the stabilizer removes species before any energy is computed: a narrow spatial packet at water’s straight arrangement, where $G_{\mathbf{y}} = \{E, (23)^*\}$, is half A_1 and half B_1 , the wavepacket form of the node rule.

8 Gauge

Splitting X into r and a shape is a choice, the *gauge*; turning the whole molecule is not one (Figure 17).

Gauge changes. Replacing $X_0(a)$ by $W(a)X_0(a)$ changes

$$r \mapsto rW(a)^{-1}, \quad R_g(a) \mapsto W(a)R_g(a)W(g \cdot a)^{-1}, \quad \psi \mapsto D^J(W)\psi.$$

The turns change by a coboundary, so (7) survives. The kinetic energy carries the mechanical connection [6, 23, 7],²⁰ which the zero-angular-momentum frame removes along a path. Gauge-invariant: the holonomy's conjugacy class (for one torsion, $e^{2\pi i \rho K}$ on the K basis), curvature invariants, the spectrum.

9 What the construction provides

The physical space is the centered configuration space $\mathcal{C}$ with its nuclear spins; the actions of $SO(3)$ and S and the Pauli rule are exact there. Reference paths, body frames, charts and gauges are choices of description, and the version model and the torsional band are reduced models. Within the stated setting the construction supplies:

- explicit configuration actions and their realizations in compatible coordinate charts (Sections 1, 4 and 5);
- spin-admissible state construction by the Pauli rule (Section 1);
- the exact species decomposition of a specified Gaussian packet, and the stabilizer-dependent exclusions for such spatial packets (Section 7);
- level counts and spin weights within the version model (Section 3);
- bounds on omitted couplings over a specified evolution time (Proposition 1);
- the reaction readout, under the stated preparation and yield assumptions (Section 3).

Preparation protocols, observable-specific signals, general nonrigid transforms and protected encodings require additional work; we do not claim a general nonrigid fiber-code or Berry-flatness theorem.

²⁰Gauge structure of this kind arises generally in simple dynamical systems, from degenerate quantum levels and in classical motion alike [24]; the one here is classical, the mechanical connection, and it enters the quantum problem through the kinetic energy.

A Proof of Proposition 1

On the version states $\mathcal{V} = \mathbb{C}[G'/H]$, equivariance makes $\langle gH | H | g'H \rangle$ a function of the double coset $Hg^{-1}g'H$, so H_0 , which keeps a set of double cosets, is equivariant too.

Blocks. A term of H_0 joins gH and $g'H$ only if $g' \in gG$, so H_0 preserves the span $\mathcal{V}_{gG}$ of the versions in each coset gG . Left multiplication by g is unitary, commutes with H_0 and maps $\mathcal{V}_G$ onto $\mathcal{V}_{gG}$, so the $[G' : G]$ blocks are copies of H_0 on $\mathcal{V}_G = \mathbb{C}[G/H]$, the cluster of G .

Levels. By Weyl's inequality each level moves by at most $\|H_1\| \leq \delta$.

Leakage. For ψ in one block and P the projector onto the others, $Pe^{-iH_0t/\hbar}\psi = 0$, so by Duhamel's formula

$$Pe^{-iHt/\hbar}\psi = -\frac{i}{\hbar} \int_0^t Pe^{-iH(t-t')/\hbar} H_1 e^{-iH_0t'/\hbar} \psi dt',$$

of norm at most $\delta t/\hbar$.

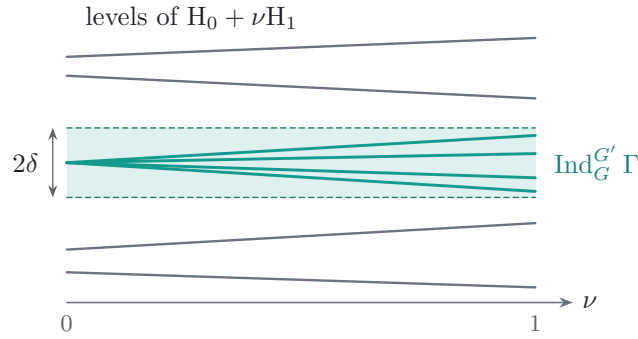


Figure 18: Representation step of the proof, schematic. Along $H_0 + \nu H_1$, $0 \leq \nu \leq 1$, levels may split inside an isolated spectral window, but the representation carried by the whole window, $\text{Ind}_G^{G'} \Gamma$, stays unchanged while its gap stays open.

Representation. An isolated level E_Γ of species Γ recurs in every block, and its eigenspace carries $\text{Ind}_G^{G'} \Gamma$. Along $H_0 + \nu H_1$, $0 \leq \nu \leq 1$ (Figure 18), its copies stay within $\nu\delta$ of E_Γ and the other levels stay more than δ away, so the spectral projector onto $[E_\Gamma - \delta, E_\Gamma + \delta]$ is continuous in ν and commutes with G' . Its irreducible multiplicities are integers varying continuously, hence constant. $\square$

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